

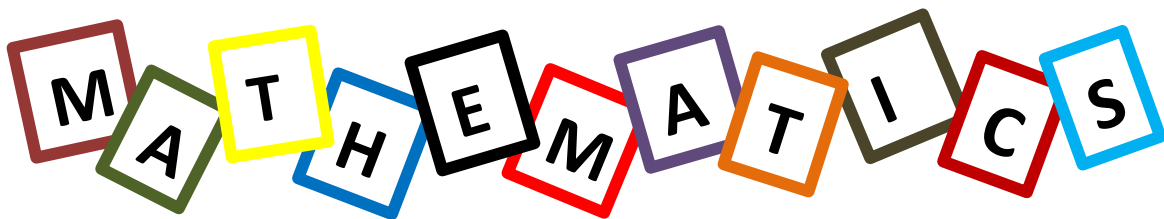


**ALPHA CHOICE INNOVATIVE ACADEMY,  
AREPO**

**GRADE 8**

**THIRD TERM**

**MASTERNOTE**



## **SYNOPSIS.**

- 1. Revision on linear graph**
- 2. Representation of real-life situation on graph**
- 3. Angles and polygons**
- 4. Angles of elevation and depression**
- 5. Bearing and distance**
- 6. Statistics**
- 7. Review and mid-term**
- 8. Statistics**
- 9. Probability**
- 10. Review of work**
- 11. Examination**
- 12. Examination**

## **. . . Weeks One - Two:**

### ***Revision and representation of real life situation on graph . . .***

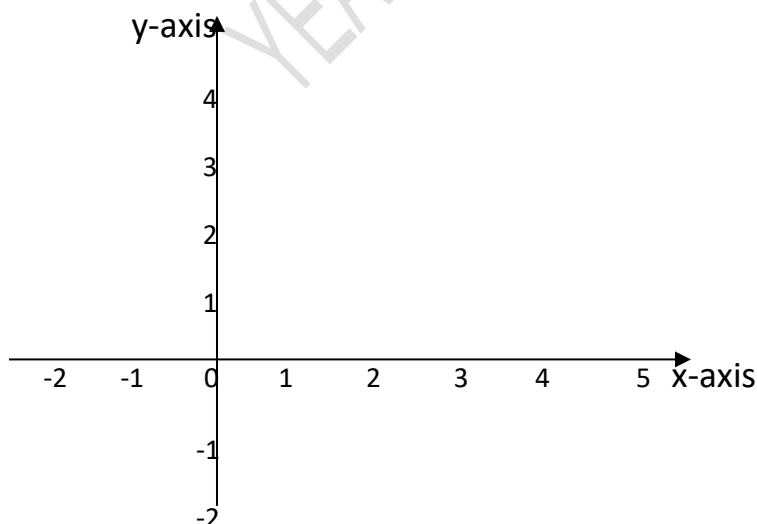
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**Behavioral objectives:** At the end of this lesson students should be able to ;

- (i) Draw a Cartesian plane and number it
- (ii) Plot points on a Cartesian plane
- (iii) Connect the points and identify the shape formed
- (iv) Construct a table of values
- (v) Choose a suitable scale
- (vi) Draw a straight line graph
- (vii) Calculate the gradient of the line
- (viii) Draw distance- time graph
- (ix) Draw velocity-time graph
- (x) Draw item- cost graph
- (xi) Re-presentation of real life situations on graphs

#### **Content**

**A Cartesian Plane** is a surface that has x-axis and y-axis.

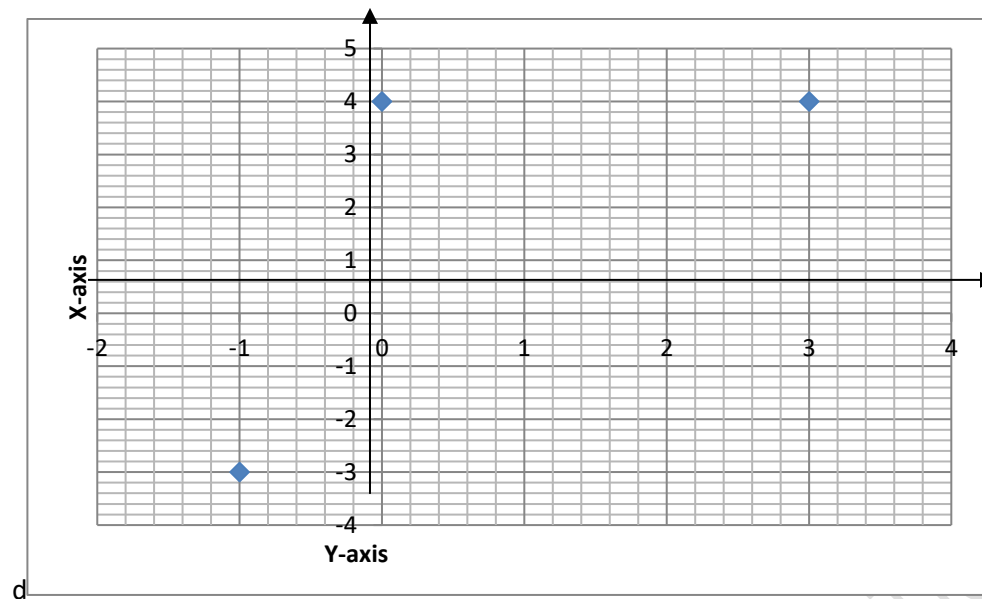


#### **Plotting of points**

To plot a point means to locate the position of the point on the Cartesian plane.

Example 1: plot the following points on a Cartesian plane. A(3, 4) B(-1, -3) C(0, 4)

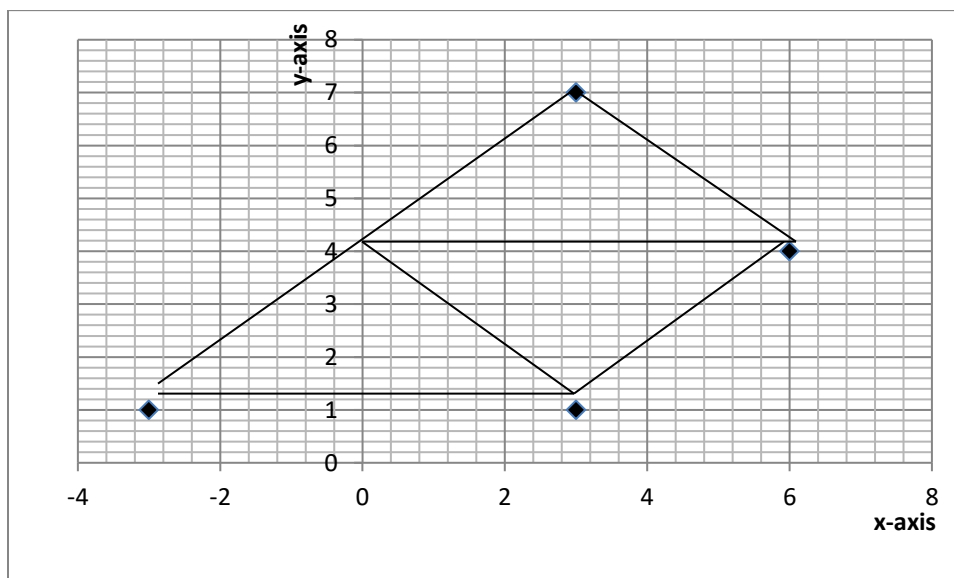
Solution



Example 2: the vertices of a quadrilateral ABCD have coordinates A(-3, 1), B(3, 1), C(6, 4), D(3, 7). E is a point (0, 4)

- Using 1cm to 1unit on both axes, plot the of points A,B,C,D and E.
- Join the vertices of quadrilateral ABCD. What kind of quadrilateral is it?
- Join CE to B and E to C. what kind of quadrilateral are
  - ABCD and
  - BCDE?

Solution



- The points are plotted above
- the quadrilateral is a trapezium
- ABCE is a parallelogram and BCDE is a square because the diagonal is equal.

### Evaluation:

- Plot the following points on a Cartesian plane. A(1, 3), B(-2, 3), C(-3, 0) and D(1, 0).
- Join the points plotted above
- What kind of shape is formed

### Assignment:

- The vertices of a triangle ABC have coordinate A(4,6), B(7,3) and C(1,3)  
Using 1cm rep. 1unit on both axes, plot the points A,B and C
- Join the vertices of the triangle ABC
- Plot another points D(3,0) and E(5,2)
- Join points C to D, D to E and E to C. what is the shape formed?

### Table of value

If 1m of wire cost ₦30, then 2m cost ₦60, 3m cost ₦90 and so on. That is (1,30); (2,60);(3,90);(4,120). These values of order pair can be shown in a table of value like this

|           |    |    |    |     |
|-----------|----|----|----|-----|
| Length(m) | 1  | 2  | 3  | 4   |
| Cost(₦)   | 30 | 60 | 90 | 120 |

Choosing scale

In drawing graphs;

- Scale is chosen for easier and accurate interpretation of the graph
- Chose a scale such that the graph paper can accommodate the plots
- Always study at the data in the table of value before choosing scale. In the table above, the graduation on the cost axis must be from ₦0 to ₦120 while on the length axis, the graduation must be from 0m to 4m
- Always choose scales of multiple 1, 2, 5 or 10. Do not choose scale of multiple of 3 and 4, its not advisable for your level now.

## Plotting of linear graph

Example 1: copy and complete the table below when TV tickets cost ₦400 each

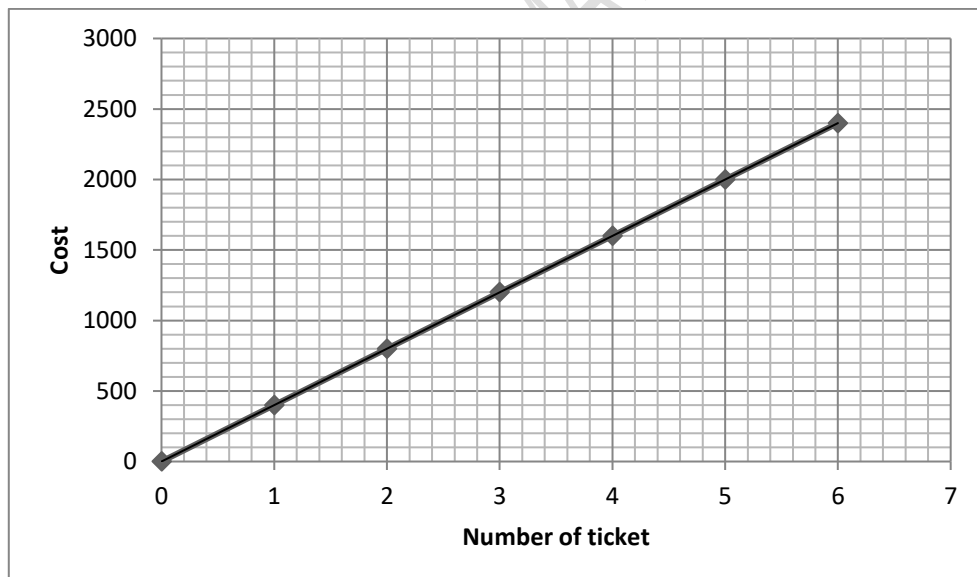
|                  |   |     |   |   |      |   |      |
|------------------|---|-----|---|---|------|---|------|
| Number of ticket | 0 | 1   | 2 | 3 | 4    | 5 | 6    |
| Cost(₦)          | 0 | 400 |   |   | 1600 |   | 2400 |

Using a scale of 1cm rep. 1 ticket on the x-axis and 2cm rep. ₦200 on the y-axis.

Draw the graph of this information.

Solution

|                  |   |     |     |      |      |      |      |
|------------------|---|-----|-----|------|------|------|------|
| Number of ticket | 0 | 1   | 2   | 3    | 4    | 5    | 6    |
| Cost(₦)          | 0 | 400 | 800 | 1200 | 1600 | 2000 | 2400 |



## Gradient

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

In the graph shown above, let the  $y_2$  be 2000 and  $y_1$  be 800 as we can see on the y-axis and the corresponding values of  $x_2$  is 5 and  $x_1$  is 2 respectively.

$$\text{The gradient of the line in the graph} = \frac{2000 - 800}{5 - 2} = \frac{1200}{3} = 400$$

**Evaluation:** A crate of 24 bottles of Fanta costs ₦1200

1. Copy and complete the table below

|                   |    |    |    |    |    |    |
|-------------------|----|----|----|----|----|----|
| Number of bottles | 10 | 20 | 30 | 40 | 50 | 60 |
| Cost(₦)           |    |    |    |    |    |    |

2. Using a scale of 2cm to 20bottles on the x-axis and 2cm to ₦1000 on y-axis.
3. Draw a graph to show this information and calculate the gradient of the line

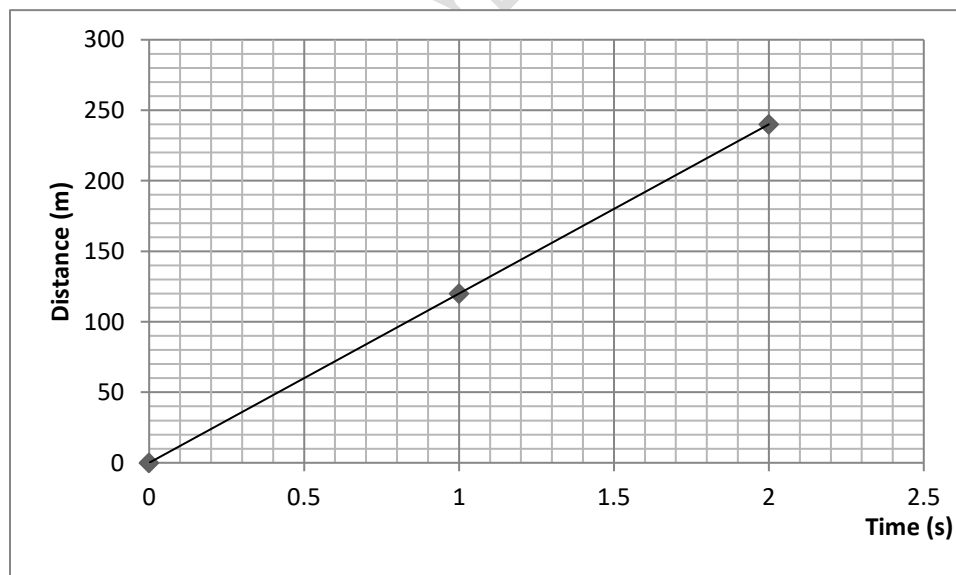
***A distance is the length between two points.***

It takes **Time** to move from a point to another. A distance time graph is the graph with x-axis represented by **Time** and Y-axis by **Distance**. This graph is used to calculate the average speed (velocity) of a man or object moving from one point to another. It can also be used to get the time spent to get to a point from a reference point.

Example 1: A boy walks from his house to a grocery store along his street for 6minutes. If the distance between the boy's house and the grocery store is 15m, use a scale of 2cm rep. 1unit on x-axis and 2cm rep. 2units on y-axis to draw a distance – time graph to show this information. Hence, find the gradient of the line

Solution

Note : the time and distance covered when he has not started walking is zero(0) so, zero is the origin of the graph



The Gradient =  $\frac{y_2 - y_1}{x_2 - x_1} = \frac{240 - 0}{2 - 0} = \frac{240}{2} = 120$

$$X_2 - x_1 \quad 6 - 0 \quad 6$$

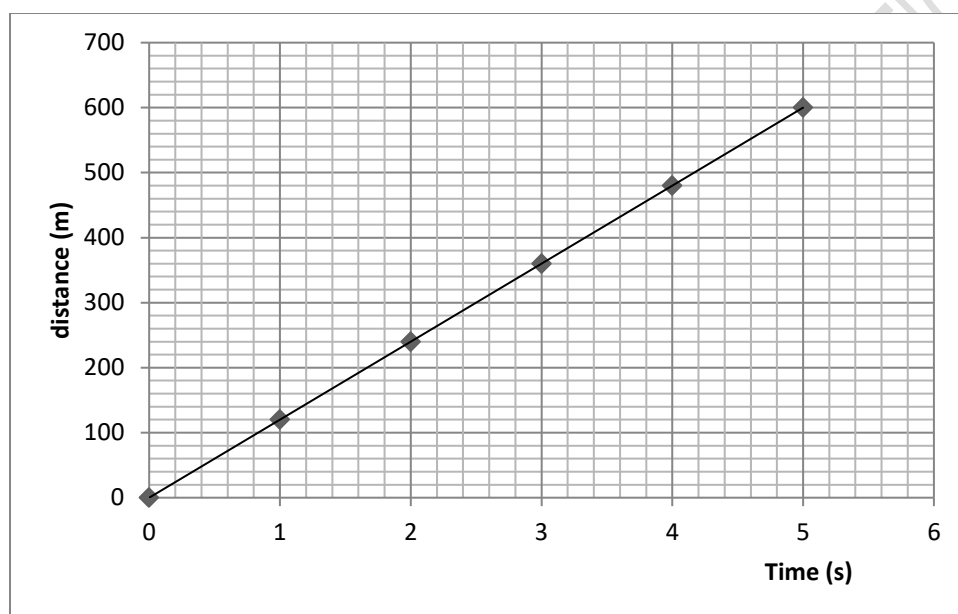
Note: the gradient of distance – time graph is known as **Speed** and the unit is meter per second, i.e. m/s

So, the average speed of the boy walking from house to store is 2.5m/s

Example 2: the table below shows the distance and time taking by a lady to cycle from college gate to her classroom block entrance. Copy the table and plot a graph to show this information. Hence find the speed at which the lady moved.

|              |   |     |     |     |     |     |
|--------------|---|-----|-----|-----|-----|-----|
| Time (min)   | 0 | 1   | 2   | 3   | 4   | 5   |
| Distance (m) | 0 | 120 | 240 | 360 | 480 | 600 |

Solution



$$\text{The Gradient} = \frac{y_2 - y_1}{X_2 - x_1} = \frac{600 - 240}{5 - 2} = \frac{360}{3} = 120$$

So, the average **Speed** of the lady is 120m/s

### Velocity – time graph

Velocity-time graph is the graph with velocity axis against time axis. It is used to calculate increase in speed at which a person or an object moves or travels with respect to time.

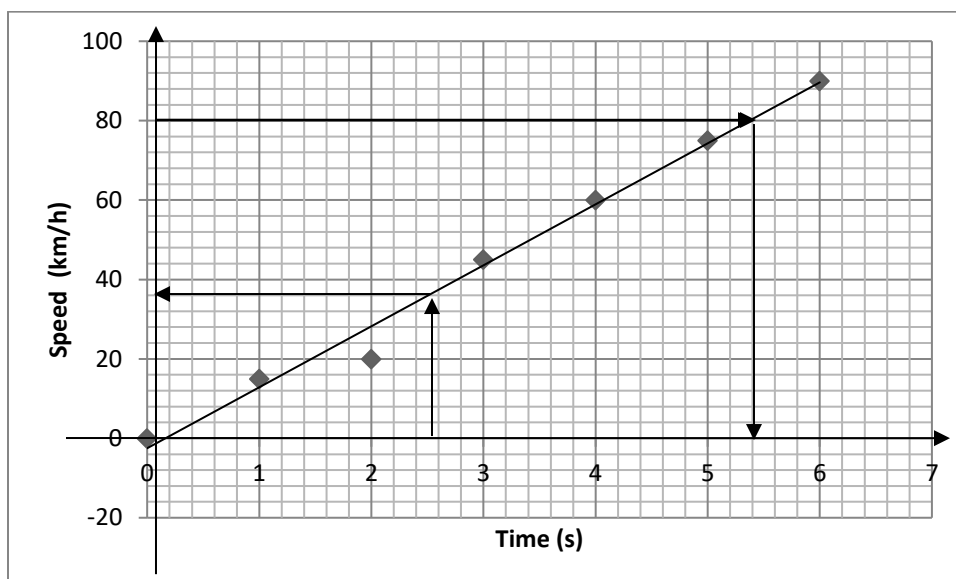
Example 3: a car increases its speed steadily over 6 seconds as shown in the table below

|             |   |    |    |    |    |    |    |
|-------------|---|----|----|----|----|----|----|
| Time (s)    | 0 | 1  | 2  | 3  | 4  | 5  | 6  |
| Speed(km/h) | 0 | 15 | 20 | 45 | 60 | 75 | 90 |

Use a scale of 2cm rep. 1 second on x-axis and 2cm rep. 10km/h on y-axis. Draw a graph of the information in the table. Hence, use your graph to find

- The speed of the car after 2.5 s
- The time to reach a speed of 80 km/h

**Solution**



- At 2.5s the speed is 36km/h
- The time to reach speed 80km/h is 5.4s

**Evaluation:**

- A girl walks along a road at a speed of 100m per minute. Copy and complete the table below

|             |   |     |     |   |   |   |   |
|-------------|---|-----|-----|---|---|---|---|
| Time (min)  | 0 | 1   | 2   | 3 | 4 | 5 | 6 |
| Distance(m) | 0 | 100 | 200 |   |   |   |   |

- Using a scale of 2cm to 1 min on the x-axis and 2cm to 100m on the y-axis, draw a graph of the information
  - Use your graph to find
    - How far the girl has walked after 5.8 min
    - How long it takes her to walk 335 m.
- The table below shows the velocity of an arrow moving in a straight line during a period of 5 seconds

|             |   |   |    |    |    |   |
|-------------|---|---|----|----|----|---|
| Time (min)  | 0 | 1 | 2  | 3  | 4  | 5 |
| Speed (m/s) | 0 | 9 | 11 | 19 | 19 | 0 |

- Choose your own scale and draw the velocity-time graph of the arrow
- Find the rate of change of velocity in the 2<sup>nd</sup> and 3<sup>rd</sup> seconds
- Find the rate of change of velocity in the last second

### Item-price graph

Example 1: 1 litre of fuel cost ₦90

- Copy and complete the table below

|              |   |    |     |     |   |   |   |
|--------------|---|----|-----|-----|---|---|---|
| Fuel(litres) | 0 | 1  | 2   | 3   | 4 | 5 | 6 |
| Cost(₦)      | 0 | 90 | 180 | 270 |   |   |   |

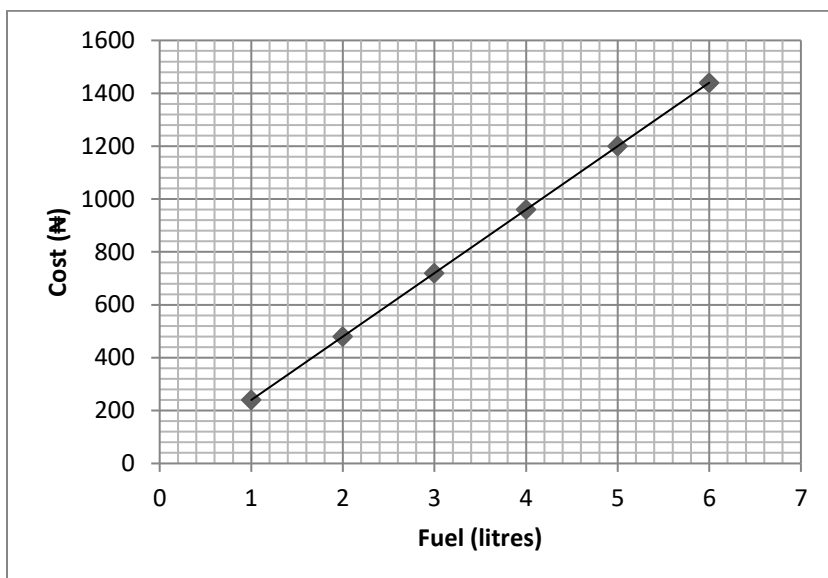
- Using a scale of 2cm to 1 litre on the horizontal axis and 2cm to ₦100 on the vertical axis, draw a graph of this information.
- Use your graph to find
  - The cost of 4.5 litres of fuel
  - How much fuel can be bought for ₦300

Solution

- 

|              |   |    |     |     |     |     |     |
|--------------|---|----|-----|-----|-----|-----|-----|
| Fuel(litres) | 0 | 1  | 2   | 3   | 4   | 5   | 6   |
| Cost(₦)      | 0 | 90 | 180 | 270 | 360 | 450 | 540 |

-



c.

i. The cost of 4.5 litres of fuel = ₦ 400

ii. How much fuel can be bought for ₦300 = 3.3 litres

### Real – life situation

Example 2: A piece of meat is taken out of a freezer. Its temperature rises steadily as shown below

|                 |    |    |    |    |    |    |
|-----------------|----|----|----|----|----|----|
| Time (min)      | 0  | 10 | 20 | 30 | 40 | 50 |
| Temperature(°C) | -6 | 0  | 6  | 12 | 18 | 24 |

a. Choose a suitable scale and draw a graph of this information

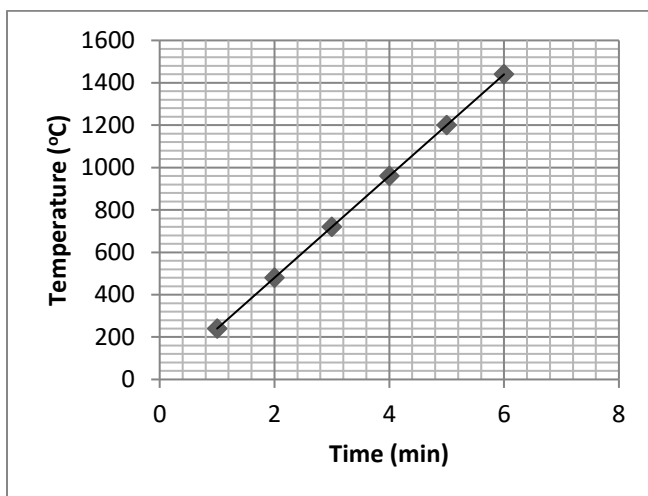
b. Use the graph to estimate

i. The temperature of the meat after 13min

ii. The time taken for the meat to reach a temperature of 20°C

Solution

a.



b.

- i. The temperature of the meat after 13 min =  $2^{\circ}\text{C}$
- ii. The time taken for the meat to reach a temperature of  $20^{\circ}\text{C}$  = 43 min

### Evaluation:

1. Sugar cost ₦240 per Kg

a. A. copy and complete the table below

|            |     |     |     |   |   |   |
|------------|-----|-----|-----|---|---|---|
| Sugar (Kg) | 1   | 2   | 3   | 4 | 5 | 6 |
| Cost (₦)   | 240 | 480 | 720 |   |   |   |

b. Using a scale of 2cm to 1 Kg on the horizontal axis and 1cm to ₦100 on the vertical axis, draw a graph of this information

c. Use your graph to find

- i. The cost of  $2\frac{1}{2}$  Kg of sugar
  - ii. How much sugar can be bought for ₦900
2. The un-stretched length of a rubber band is 120mm. when masses were hung on the rubber band, its total length changed as given below

|            |     |     |     |     |     |      |
|------------|-----|-----|-----|-----|-----|------|
| Mass (g)   | 0   | 200 | 400 | 600 | 800 | 1000 |
| length(mm) | 120 | 170 | 220 | 270 | 320 | 370  |

a. Choose a suitable scale and draw a graph of data

b. Use your graph to find:

- i. The length of the rubber band when a mass of 250 g is hung on it
- ii. The mass which stretches the rubber and to a length of 300mm.

**Assignment:** Essential Mathematics, page 182. Exercise 16.1, No1 (a,b,c),2.

Exercise 16.2, No 2,3

Exercise 16.3, No 1,2.

Additional questions.....

1) The Cartesian Plane is made up of the X and y axis?

answer choices

A.True

B.False

2) The X axis is the;

A.Vertical Line

B.Diagonal Line

C.Horizontal lines

D.All the lines

3) The y-axis is the vertical line on the Cartesian Plane?

A.True

B.False

4) The Cartesian Plane is compose of how many quadrants

A.0

B.2

C.4

D.6

5) An ordered pair consist of what to values on the Cartesian Plane?

A.(x,y)

B.(y,x)

C.Both

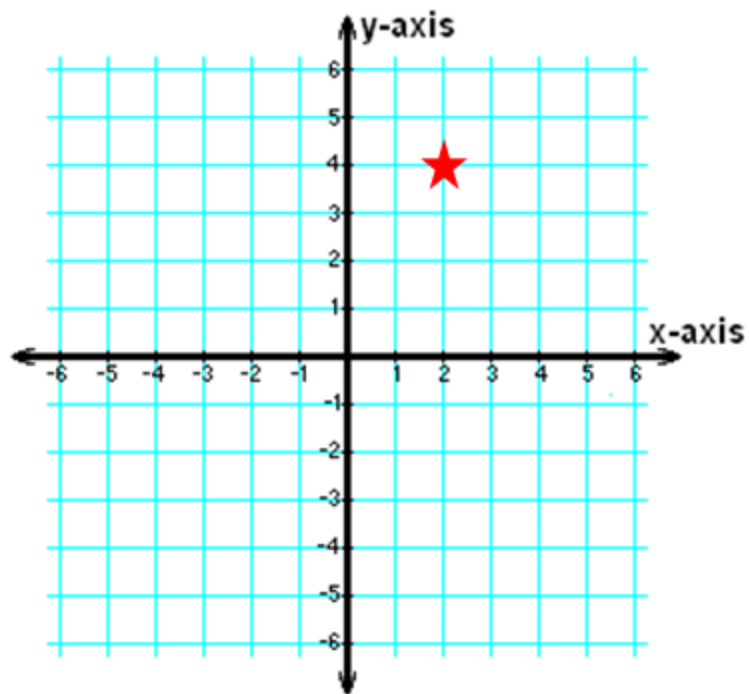
D.Neither

6. Which number in the ordered pair,  $(-4, 5)$  represents the x-value?

A.-4

B.5

7. Select the correct coordinates for the star.



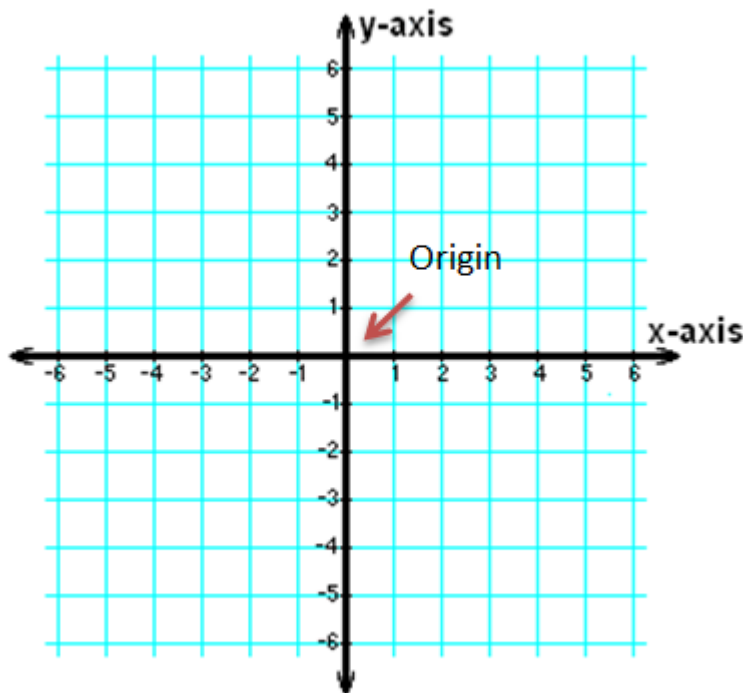
☐ (4,2)

☐ (2,4)

☐ (2,-4)

☐ (2,5)

8. The arrow points to the origin on the graph. What are the coordinates for the origin?



9. Graph the following ordered pairs in the cartesian plane:

1. (7, 4)
2. (-5, 2)

10. Construct a graph of  $Y = 5x - 3$  on the graph, taking  $x$  from -2 to 3

## **WEEK 3:**

### ***Angles and polygon . . .***

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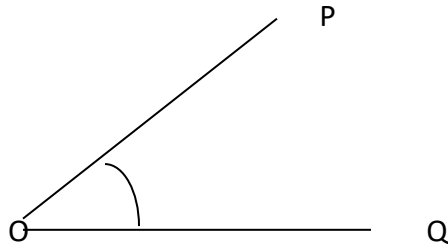
**Objectives:** At the end of this lesson students should be able to ;

- i. Define angle and polygon
- ii. Classify angles
- iii. Identify the properties of angles
- iv. Mention and define types of polygon
- v. Use the formula to find the sum of interior angles in a polygon

vi. Find how many sides has a polygon of a given sum of the angles.

**Content:**

An angle is the amount of distance between the directions of two lines or surface where they meet.



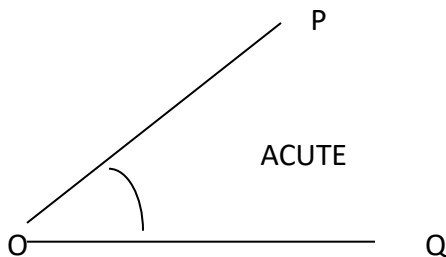
OP and OQ are two lines, which meet at O. the space formed; when OP turns to meet OQ, the direction of the movement is clockwise.

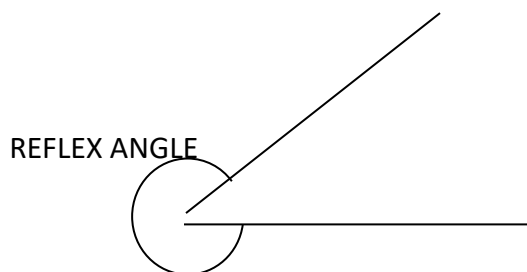
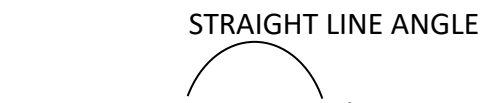
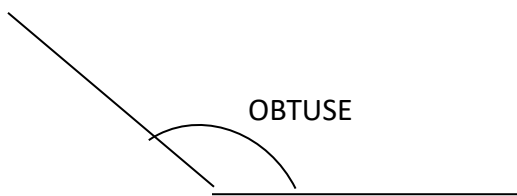
The point O where the lines meet is called the vertex. OP and OQ are called the arms of the angle.

**Classification of angles**

Angles are classified as follows;

- a. Acute : the angle that is less than  $90^\circ$
- b. Right angle: the angle that equals to  $90^\circ$
- c. Obtuse: the angle that is greater than  $90^\circ$
- d. Straight line angle: the angle that is half of a turn' it is equal to  $180^\circ$
- e. Reflex : the angle that is greater than  $180^\circ$  but less than  $360^\circ$





## POLYGON

A polygon is a plane closed shape with three or more angles and edges

The table below shows some types of the polygon, number of sides, number of angles and number of triangles

| Polygon        | Number of sides | Number of angles | Number of triangles |
|----------------|-----------------|------------------|---------------------|
| Triangle       | 3               | 3                | 1                   |
| Quadrilaterals | 4               | 4                | 2                   |
| Pentagon       | 5               | 5                | 3                   |
| Hexagon        | 6               | 6                | 4                   |
| Heptagon       | 7               | 7                | 5                   |
| Octagon        | 8               | 8                | 6                   |
| Nonagon        | 9               | 9                | 7                   |
| Decagon        | 10              | 10               | 8                   |

From the table above, we deduce that the sum of interior angles in a polygon is  $S = (n - 2)180^\circ$ , where  $n$  represents the number of sides.

The table above shows that

- (1) A triangle is a three-sided polygon
- (2) Quadrilaterals e.g. square, rectangle, trapezium e.t.c. are four-sided polygons.

**Evaluation:**

1. define angle and polygon
2. which angle has  $47^\circ$
3. what is the name of the angle that is greater than  $180^\circ$  but less than  $360^\circ$
4. differentiate between obtuse angle and reflex angle
5. what is the name of the shape with the following number of sides (a) 5 , (b) 7, (c) 10
6. mention 8 different types of polygon

**Assignment:** N.G.M. page 196 Ex 19 No 6 – 10

### **SUM OF INTERIOR ANGLES IN A POLYGON**

The formula for finding the sum of interior angles of a regular polygon =  $(n - 2)180^\circ$

Example 1: calculate the sum of the interior angles of a regular hexagon

Solution

A regular hexagon has six sides i.e.  $n = 6$

$$. S = (n - 2)180^\circ$$

$$= (6 - 2) \times 180 = 4 \times 180 = 720^\circ$$

Example 2: find the value of each angle of a regular hexagon

Solution

From example 1 above, the sum of angles in a hexagon is  $720^\circ$ . there are six angles in hexagon

$$\therefore \text{Each angle} = \frac{720}{6} = 120^\circ$$

Example 3: find the number of sides in a polygon if the sum of angles is  $2880^\circ$

Solution

$$\text{Sum of all angles in a polygon} = S = (n - 2)180^\circ$$

$$\therefore S = (n - 2)180^\circ = 2880$$

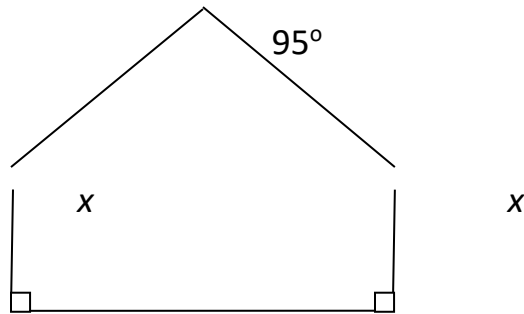
Divide both sides by  $180^\circ$

$$\frac{(n-2)180^\circ}{180} = \frac{2880}{180}$$

$$(n - 2) = 16 \quad \text{i.e. } n = 18.$$

The polygon has 18 sides

Example 4: find  $x$  and  $y$  in the shape shown below



**Solution**

The shape above is an irregular pentagon,

The sum of angles in a pentagon  $S = (5 - 2)180^\circ = 540^\circ$

The given angles are  $90^\circ$ ,  $90^\circ$ , and  $95^\circ$

$$\therefore 90^\circ + 90^\circ + 95^\circ + x + x = 540^\circ$$

$$275 + 2x = 540^\circ$$

$$2x = 540 - 275^\circ = 265^\circ$$

$$x = \frac{265}{2} = 132.5^\circ$$

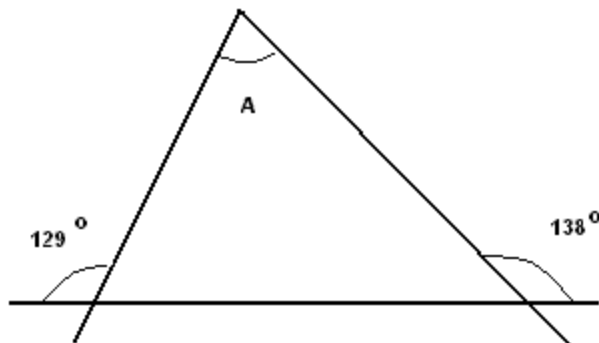
**Evaluation:**

1. Find the sum of angles in a decagon
2. Find the angles in a regular polygon of sum  $900^\circ$
3. The sum of the angles in a nonagon is  $1260^\circ$ . if the sum of six of the angles is  $840^\circ$ . calculate the size of the other three equal angles

**Assignment:** N.G.M. page 198 Ex 14D, No 4, 5 and 9

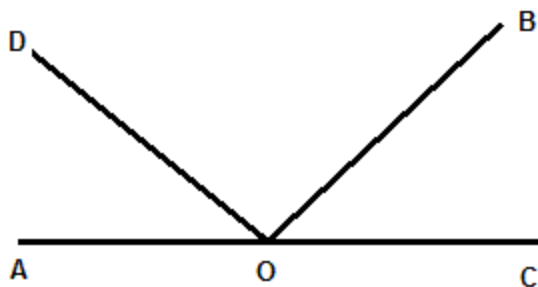
ADDITIONAL QUESTIONS.....

- 1) Find the measure of each interior angle of a regular polygon having 10 sides
- 2) Is it possible to have a regular polygon each of whose interior angles is  $100^\circ$ ?
- 3) What is the sum of all interior angles of a regular pentagon
- 4) What is the sum of all interior angles of a regular hexagon
- 5) What is the sum of all interior angles of a regular octagon
- 6) Find the measure of angle A in the figure below



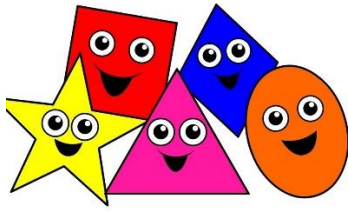
[www.analyzemath.com](http://www.analyzemath.com)

- 7) The size of angle AOB is equal to 132 degrees and the size of angle COD is equal to 141 degrees. Find the size of angle DOB.



[www.analyzemath.com](http://www.analyzemath.com)

- 8) Which of the following is not a polygon?
- A. trapezoid
  - B. semicircle
  - C. rhombus
  - D. octagon



9) A POLYGON IS:

A figure with more than 2 sides

A.correct

B.incorrect

C.not sure

10) How many sides does a quadrilateral have?

A.5

B.3

C.2

D.4

YEAR 8 MATHS

## **... Week Four : Angle of elevation and depression ...**

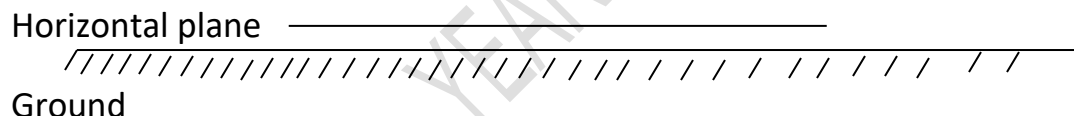
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**Behavioral objectives:** At the end of this lesson students should be able to ;

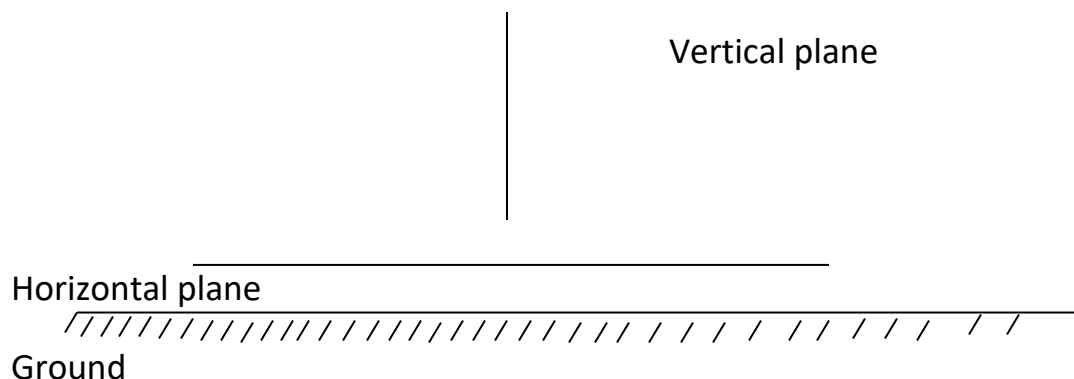
- i. Recognize the horizontal plane and vertical plane
- ii. Define angle of elevation
- iii. Give examples of angles of elevation
- iv. State the uses of angles of elevation and Pythagoras theorem
- v. Use ruler, set square and protractor to take measurement
- vi. Calculate height and distance between two points
- vii. Define angle of depression
- viii. Discuss the relationship between angle of elevation and depression
- ix. Calculate distance between two points

### **Content:**

A horizontal plane is the plane parallel to the surface of the Earth



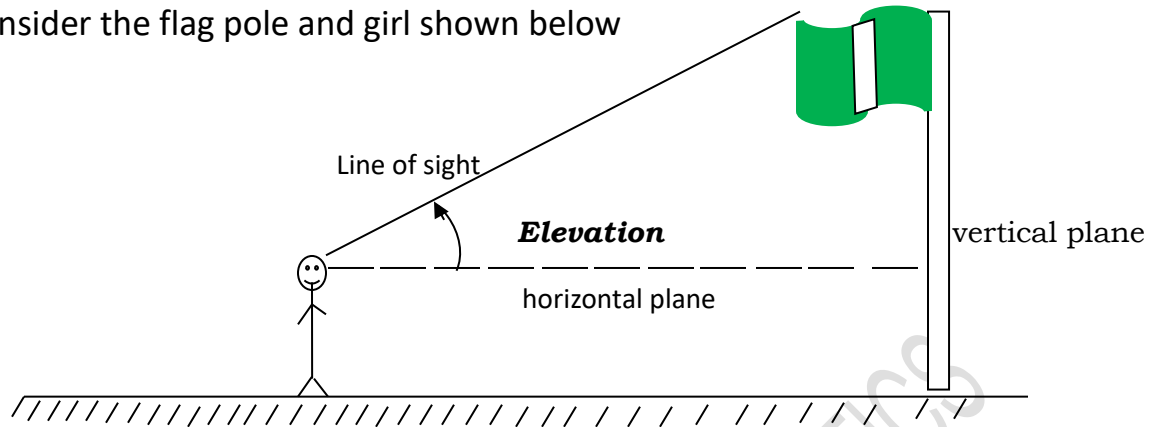
A vertical plane is the plane perpendicular to the horizontal plane



## Angle of elevation ( $\alpha$ )

An angle of elevation of an object from an observer is the angle between the observer's line of sight and the horizontal.

Consider the flag pole and girl shown below



## Examples of angles of elevation

from an observer at a lower level, we have angles of elevation of ;

1. The sun and the top of
2. A tree
3. A tower

## Uses of angles of elevation

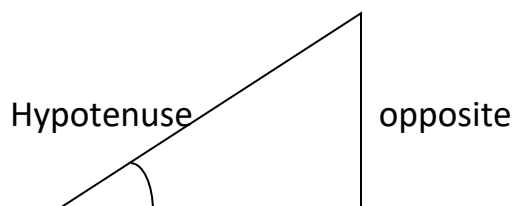
Angles of elevation can be used to find;

- i. Heights of tall objects, which are not easy to measure physically
- ii. Distances on the ground between two points which cannot be measured because of observation between the points

N.B: A Clinometer is an instrument which is used to measure angle of elevation

## Uses of Pythagora's Rule

Using any Right angle triangle



▼  $\alpha$

Adjacent

Hypotenuse = Hyp

Adjacent = Adj

Opposite = Opp

Pythagoras state that

$$\text{Hyp}^2 = \text{Opp}^2 + \text{adj}^2$$

Hypotenuse = line of sight

Opposite = vertical

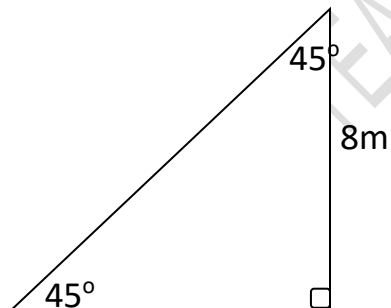
Adjacent = Horizontal

Calculation:

The angle of elevation of the top of a tree from a point on the ground is  $45^\circ$ . If the tree is 8m high, how far is the point from the base of the tree? (use 1cm rep 1m)

Solution

Using a ruler, protractor and  $45^\circ$  set square. Draw a 8cm vertical line



BC is the tree

A is the observer and  $\angle CAB = 45^\circ$  (angle of elevation)

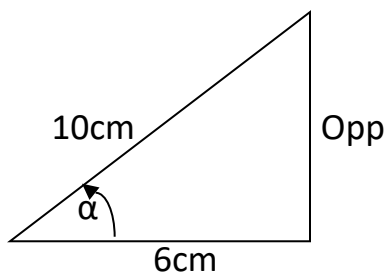
Also,  $\angle ACB = 45^\circ$  (sum of angles in a triangle)

The triangle is an isosceles triangle

Distance =  $AB = BC = 8\text{m}$

### Evaluation:

1. Define horizontal line, vertical line and angle of elevation
2. Give 5 examples of angle of elevation from an observer
3. State the uses of angle of elevation
4. State Pythagora's Theorem
5. Solve the angle in the triangle below



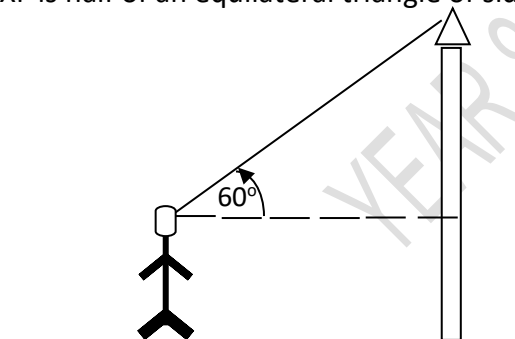
**Assignment:** N.G.M. page 127, Ex 7b, no. 5 - 9

### HEIGHT AND DISTANCE BETWEEN TWO POINTS

Example 1: The angle of elevation of the top X of a vertical pole from a point P on a level ground is  $60^\circ$ . The distance from P to the foot of the pole is 15m. Find the height of the pole.

Solution

$\triangle$  PXF is half of an equilateral triangle of side XP and height XF



$$\text{Hypotenuse} = 2 \times \text{adjacent} = 2 \times 15\text{m} = 30\text{m}$$

By Pythagora's Theorem

$$\text{Hyp}^2 = \text{opp}^2 + \text{adj}^2$$

$$\text{Hyp} = 30\text{m}, \quad \text{opp} = h, \quad \text{adj} = 15\text{m}$$

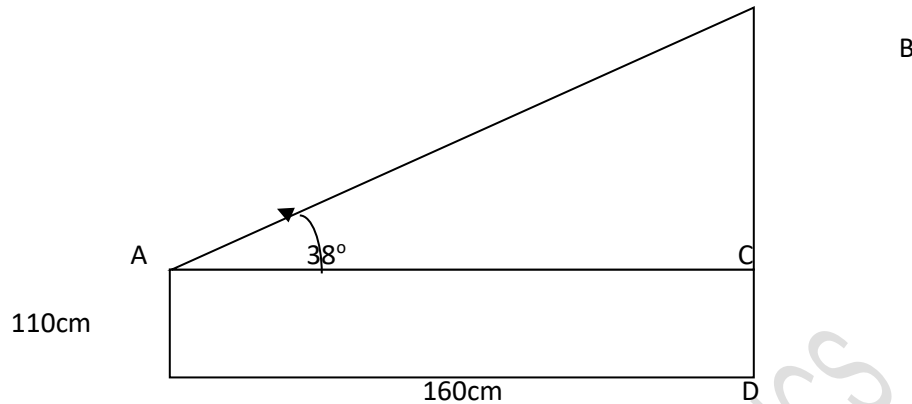
$$30^2 = h^2 + 15^2$$

$$h^2 = 30^2 - 15^2 = 900 - 225 = 675$$

$$h = \sqrt{675} = 26\text{m}$$

Example 2: If a girl is 110cm and she is standing 160cm away from a flagpole. If the angle of her line of sight made with the flying flag is  $38^\circ$ . Using ruler and protractor for measurement and scale drawing of ratio 1cm to 20cm, calculate the height of the flag pole in meter.

Solution

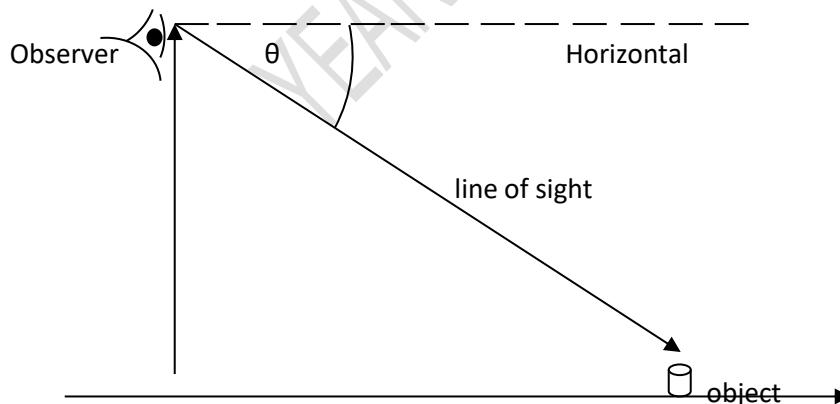


The scale diagram shown the measurement of BC is 6.0cm, which equals to  $(6 \times 20)\text{cm} = 120\text{cm}$ .

The height of the flag pole =  $120\text{cm} + 110\text{cm} = 230\text{cm} = 2.3\text{m}$

### Angle of Depression

An angle of depression of an object from an observer is the angle between the observer's line of sight and the horizontal. It is denoted by  $\theta$



Example of angles of depression

Angle of depression of;

- i. Control tower from an aero plane
- ii. A boat from the top of a vertical cliff
- iii. Object on the ground from the top of

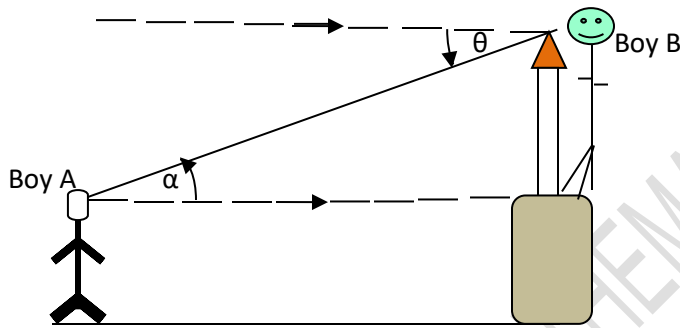
- a. A building
- b. A tall tree

Uses of angle of depression

- a.) To find the height of tall objects, without measuring them physically
- b.) To find the distance between two points on the same horizontal plane

Relationship Between the Angle Of Elevation And Depression

The diagram below shows both the angle of elevation and depression with two boys at different positions



Boy A makes angle  $\alpha$  with boy B (i.e. angle of elevation of Boy B from Boy A).  
likewise Boy B makes angle  $\theta$  with Boy A (i.e. the angle of depression of Boy A from Boy B).

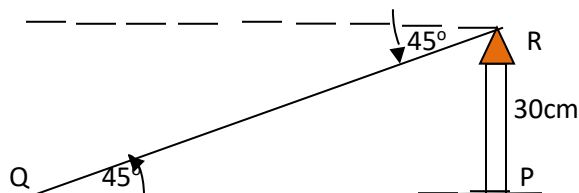
These angles are alternate to each other;

So, angle A = angle B i.e.  $\alpha = \theta$

$\therefore$  The angle of elevation is equal to the angle of depression

Example : the angle of depression of a point Q from a vertical tower PR, 30m high, is  $45^\circ$ . if the foot P of the tower is on the same horizontal plane as Q. find PQ

Solution

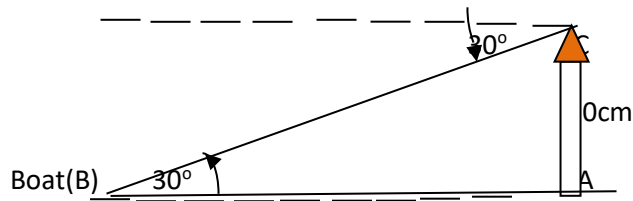


$\Delta PQR$  is an isosceles right-angled triangle

$$IPQI = IPRI = 30m$$

Example 2: the angle of depression of a boat from the top of a cliff 20m high is  $30^\circ$ .  
How far is the boat from the foot of the cliff?

Solution



$\triangle ABC$  is half of an equilateral triangle of side AB and height AC

$$IBC I = 2 IAC I = 2(20) \text{ cm} = 40\text{cm}$$

By Pythagora's Rule

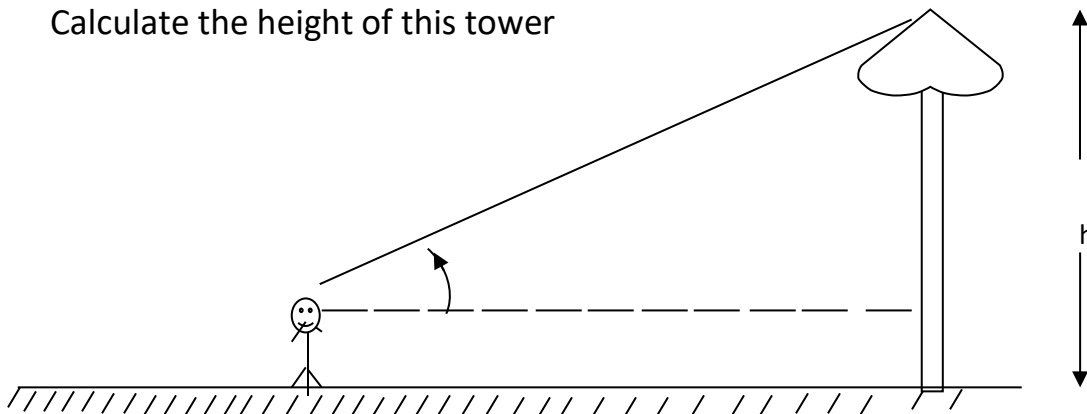
$$IABI^2 + IAC I^2 = IBC I^2 \text{ i.e. } IABI = 40^2 - 20^2 = 1600 - 400 = 1200$$

$$IABI = \sqrt{1200} = 20\sqrt{3} \text{ cm}$$

The boat is  $20\sqrt{3}$  cm from the foot of the cliff.

**Evaluation:**

- Find  $h$  in the diagram shown below. Use a ruler, protractor and pencil.  
Calculate the height of this tower



2. The angle of elevation of a building from a measuring instrument placed on the ground is  $30^\circ$ . If the building is 40m high, how far is the instrument from the foot of the building?
3. A man 1.7m tall observes a bird on top of a tree at an angle of  $30^\circ$ . If the distance between the man's head and bird is 25m, what is the height of the tree?
4. A cliff on the bank of a river is 300m high. If the angle of depression of a point on the opposite side of the river is  $45^\circ$ , find the width of the river.

**Assignment:** page 129, Ex 3C, no. 4 -8

#### ADDITIONAL QUESTIONS.....

- 1) If a person stands and looks up at an object, the angle of ----- is the angle between the horizontal line of sight and the object.
- 2) From the top of a cliff of 200 m observes that the angle of depression of a high, Dele boat at sea is  $35^\circ$ . Find the distance between the boat and the foot of the cliff.
- 3) When the altitude of the sun is  $50^\circ$ , find the length of the shadow cast by a vertical pole of height 30 m on the horizontal ground.
- 4) **You stand 25 m from the base of a tree and the angle of elevation from the ground to the top of the tree is 48 degrees. How tall is the tree?**



27.8 m



18.6 m



16.7 m



0.04 m

5) You look up at a 62 m tall building and find that the angle of elevation from the ground to the top of the building is 51 degrees. How far away from the building's base are you?

- a. 76.6 m
- b. 50.2 m
- c. 48.2 m
- d. 39.0 m

6) Joe stands at the top of a cliff and looks down at a 49-degree angle of depression at a boulder that is 17 m away from the bottom of the cliff. How tall is the cliff?

- a. 19.6 m
- b. 14.8 m
- c. 11.2 m
- d. 25.9 m

7) If a person stands and looks down at an object, the angle of ----- is the angle between the horizontal line of sight and the object.

8) A girl with eyes-level height of 2.0m observes that the angle of elevation of the top of a tower 30 m away is  $60^\circ$ . Calculate the height of the tower.

9) A man is 1.8 m tall. He stands 50 m away from the base of a building. His angle of elevation to the top of the building is  $70^\circ$ .

Calculate an estimate of the height of the building. Give your answer to an appropriate degree of accuracy.

10) An airplane takes off 600 feet in front of a 60 foot building. At what angle of elevation must the plane take off in order to avoid crashing into the building?

## ... Week Five : Distances and Bearing ...

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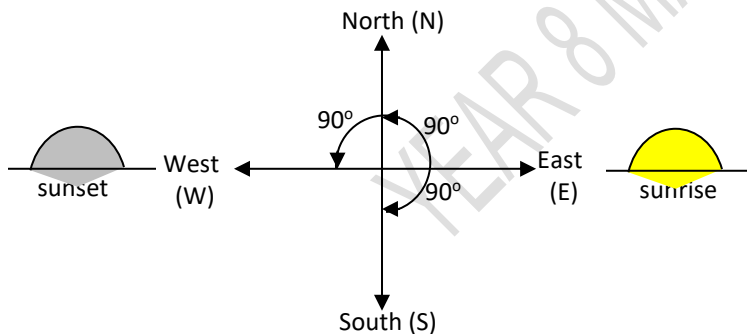
**Behavioral objectives:** At the end of this lesson students should be able to ;

1. Identify and name the four main compass direction and the four secondary directions
2. State types of bearing
3. Find the bearing of one point from another using the three different types of bearing
4. Calculate three figure bearing

### Finding Direction

The four main directions are : North, South, East and West

Compass is a tool we use to find direction of a place from a particular point or place.



### Points on Compass

- i. North
- ii. South
- iii. East
- iv. West
- v. North west (NW) i.e  $N45^{\circ}W$
- vi. North east (NE) i.e  $N45^{\circ}E$
- vii. South west (SW) i.e  $S45^{\circ}W$

viii. South east (SE ) i.e  $S45^{\circ}E$

Note : The angle between N and NE is  $45^{\circ}$  and so on.

### Activity

(1) Do the following and state the final direction that you face

- Face East, turn through an angle of  $180^{\circ}$
- Face East, turn through  $45^{\circ}$  to your left
- Face East, turn through  $90^{\circ}$  to your right
- Face East, turn through  $90^{\circ}$  to your left

(2) What is the angle between the following directions

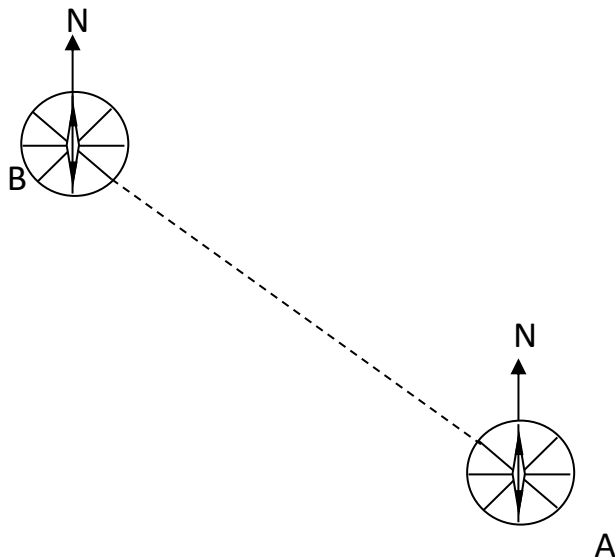
- N and S
- NE and E
- W and E
- E and SW
- NW and SE

### Bearing is the direction between two points

If Temmy is standing 4m north of Fauzat. We say that Temilola is north of Fauzat. That is, the direction(bearing) of Temmy from Fauzat is north. Also, the bearing of Fauzat from Temmy is south

Compass bearing

The diagrams below show the direction of A from B and vice versa



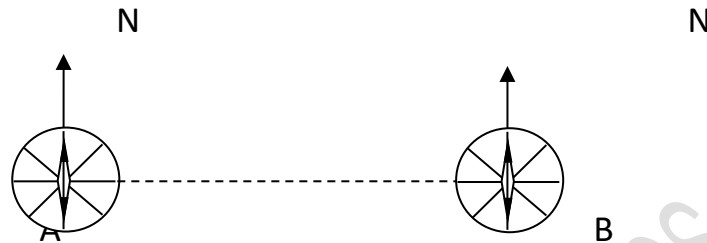
The pointer of both compasses point northwards (N). the compass at B is in a direction NW of A. therefore the bearing of B from A is NW. Similarly, the bearing of A from B is SE.

### Activity

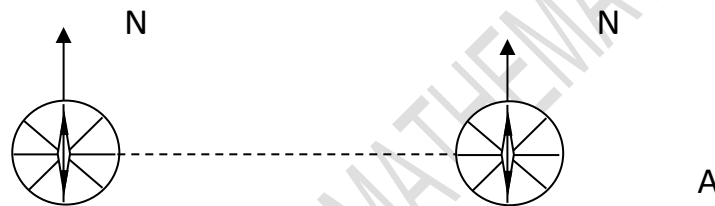
In each part in the diagram below state

- a. The bearing of B from A
- b. Bearing of A from B

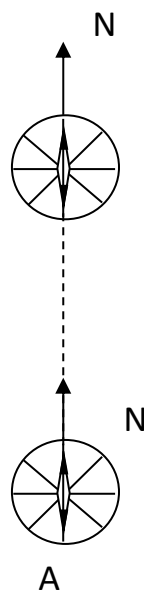
1.



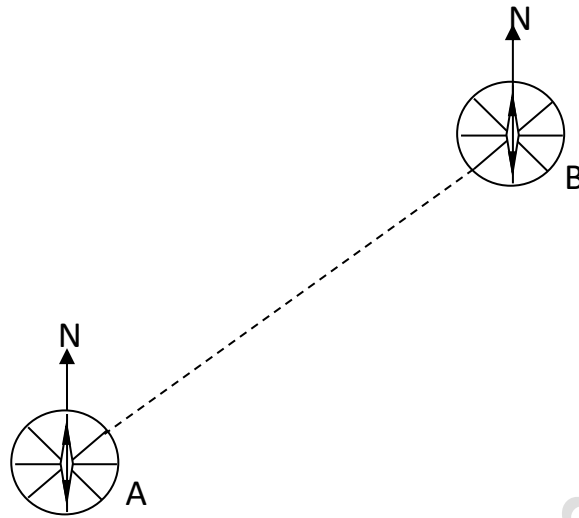
2.



3.



4.

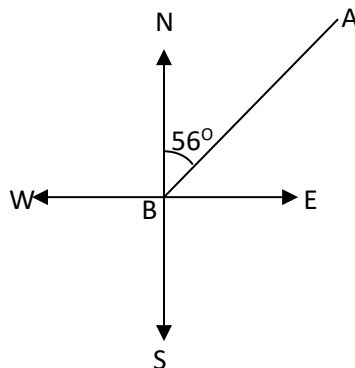


### Types of bearing

1. Compass bearing
2. Acute angle bearing e.g  $N30^{\circ}W$ ,  $S24^{\circ}E$ ,  $S87^{\circ}W$ , e.t.c.
3. Three-figure bearing (true bearing) e.g  $023^{\circ}$ ,  $089^{\circ}$ ,  $100^{\circ}$ ,  $177^{\circ}$ ,  $206^{\circ}$ ,  $333^{\circ}$ , e.t.c. The reading starts clock-wisely from the North only.

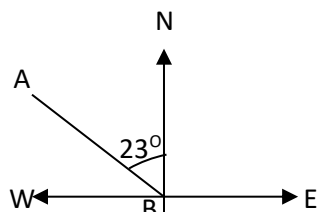
**Activity 3** : State the bearing of A from B in the diagram below using

- a. Acute angle bearing
  - b. True bearing
- 1.



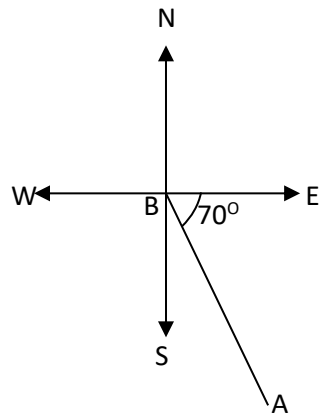
|                        |                         |
|------------------------|-------------------------|
| Acute angle<br>bearing | Three figure<br>bearing |
|------------------------|-------------------------|

2.



|                        |                         |
|------------------------|-------------------------|
| Acute angle<br>bearing | Three figure<br>bearing |
|------------------------|-------------------------|

3.



|                        |                         |
|------------------------|-------------------------|
| Acute angle<br>bearing | Three figure<br>bearing |
|------------------------|-------------------------|

**EVALUATION:** N.G.M. JSS2 Page 228 – 229, Ex. 24e

1. No. 1 a – h
2. No. 2
3. No. 3

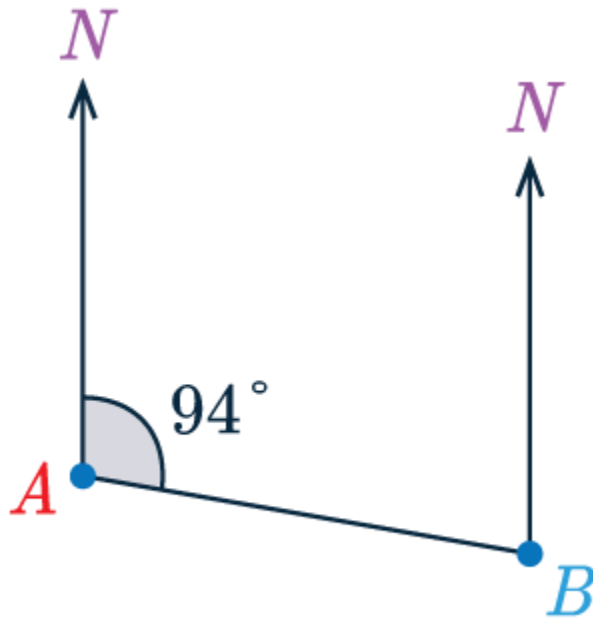
**ASSIGNMENT:** N.G.M. JSS2 Page 228 – 229, Ex. 24e

1. No. 4 a – e
2. No. 5 a – g

Additional questions.....

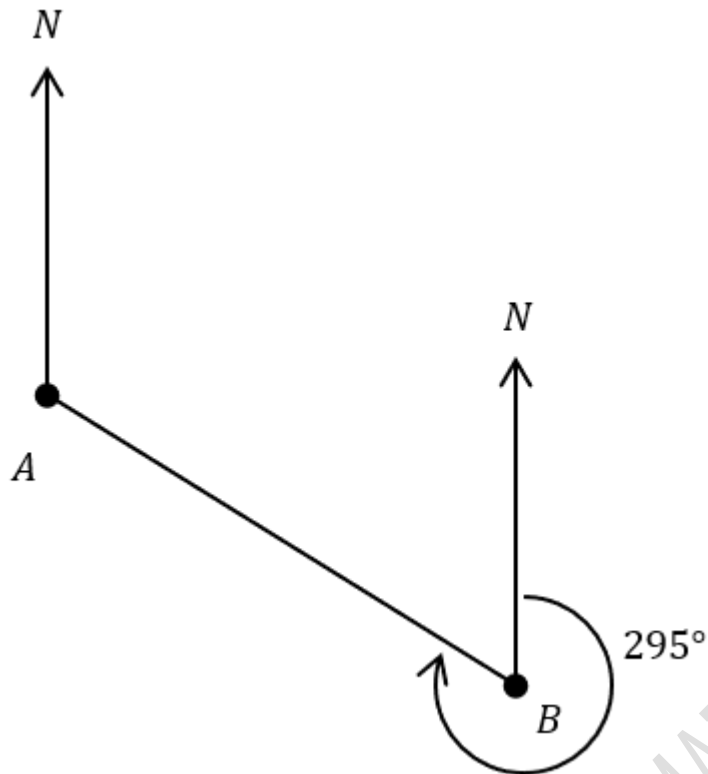
- 1) Write down the three-figure bearings of each of the following directions: (a) N  $25^\circ$  E (b) S  $42^\circ$  E
- 2) A ship is 4 km due north and then 3 km on a bearing of  $60^\circ$ . Find the distance from his original position.
- 3) A village is 10 km on a bearing  $050^\circ$  from a point O. How far is the village North of O?
- 4) The angle of elevation of the top of a building from a point 80 m away on a level ground is  $25^\circ$ .  
Calculate the height of the building to the nearest metre.

- 5) A ship leaves a port at noon and has a bearing of  $207^\circ$ . If the ship is traveling at 20 miles per hour, how many miles south and how many miles west is the ship from its departure point at 6 pm?
- 6) Find the bearing of A from B



Not drawn accurately

- 7) The diagram below shows the bearing of A from B. Find the bearing of B from A.



- A. 115
- B. 225
- C. 105

8) What is the total displacement of a student who walks 3 blocks east, 2 blocks north, 1 block west, and then 2 blocks south?

a. A.

2 blocks east

b. B.

2 blocks west

c. C.

8 blocks

d. D.

9) A girl leaves a history classroom and walks 10. meters north to a drinking fountain. Then she turns and walks 30. meters south to an art classroom. What is the girl's total displacement from the history classroom to the art classroom?

o A.

20. m south

☐ B.

20. m north

☐ C.

40. m north

☐ D.

40. m south

10) A student walks 1.0 kilometer due east and 1.0 kilometer due south. Then she runs 2.0 kilometers due west. The magnitude of the student's resultant displacement is close to?

YEAR 8 MATHEMATICS

### ***. . . Weeks Six and Eight : Statistics . . .***

---

**Behavioral objectives:** At the end of this lesson students should be able to collect data from different places or areas and present it in;

- (i) Ordered form
- (ii) Frequency form
- (iii) Construct a pie chart

- (iv) Use fraction of angle in a circle to find each angle of the corresponding frequency of the variable
- (v) Draw an appropriate size of circle, measure different angles on the circle with protractor and draw the angles
- (vi) Read the information on a given Pie chart and use it to solve a simple problem.
- (vii) Use fraction of angle in a circle to find each angle of the corresponding frequency of the variable
- (viii) Draw an appropriate size of circle, measure different angles on the circle with protractor and draw the angles
- (ix) Read the information on a given Pie chart and use it to solve a simple problem.
- (x) Solve more problem using Pie chart
- (xi) State the usefulness of data for statistical purpose

### **Content:**

Activity 1: To be done by the students and supervised by the teacher

- 1.) Collect the ages of all the children in your classroom.
- 2.) Arrange the numbers from the smallest to the highest. From the arrangement you will discover the numbers of younger students, the older students and the students whose ages are in between.

Activity 2:

- 1.) Collect the mathematics scores of all students in your school
- 2.) Arrange the numbers from the smallest to the highest. From the arrangement you will discover the numbers of younger students, the older students and the students whose ages are in between.

Activity 3: Collect the shoe sizes of all students in your school and present your result on frequency table. For Example,

|                              |      |      |    |    |    |      |    |    |
|------------------------------|------|------|----|----|----|------|----|----|
| Shirt size                   | 27 ½ | 28 ¾ | 31 | 32 | 36 | 38 ½ | 39 | 40 |
| Frequency(number of student) | 18   | 14   | 10 | 27 | 22 | 32   | 11 | 3  |

### **Evaluation:**

- 1.) Collect the ages of all the children in your classroom. Arrange the numbers from the smallest to the highest
- 2.) Collect the mathematics scores of all students in your school. Arrange the numbers from the highest to the smallest
- 3.) Collect the shoe sizes of all students in your school and present your result on frequency table

**Assignment:** page 189 Ex. 12. No. 2-3

## **PIE-CHART**

Pie- chart is a circular form of data presentation. The pie chart is divided into sectors.

We know the sum of angles in a circle is  $360^\circ$ . Each sector will be calculated based on the following steps:

1. Add up the data and get the total number (i.e. N)
2. Divide each mark or data by the total and multiply by  $360^\circ$ . This gives us the angle of each sector.
3. Angle =  $\frac{\text{Number in category}}{\text{Total number (frequency)}} \times 360^\circ$
4. When all the angles are added up, it must be equal to  $360^\circ$ .
5. Draw your circle, use the protractor to measure each angle.

Example 1: The table below shows the grades of some students in the mathematics NECO/JSSCE examination in the year 2004.

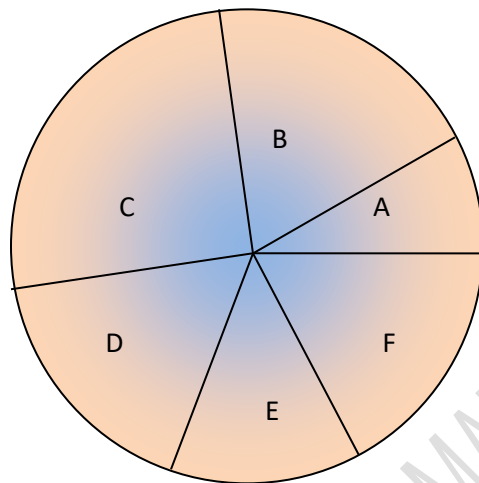
| Grade           | A  | B  | C  | D  | E  | F  |
|-----------------|----|----|----|----|----|----|
| No. of students | 10 | 20 | 32 | 25 | 15 | 18 |

Show this information on Pie-chart.

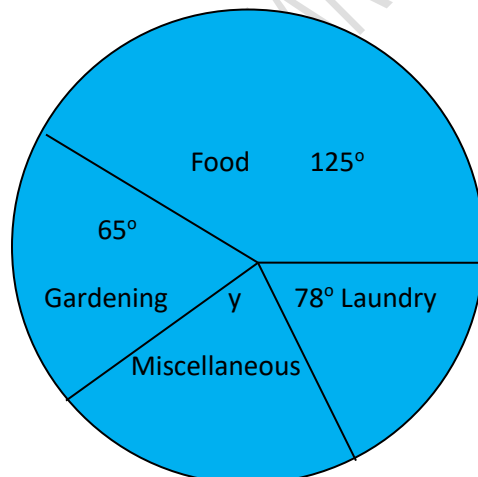
### Solution

Total no of students =  $10+20+32+25+15+ 18 = 120$  students

| Grade | No. of students | Angles of sectors                            |
|-------|-----------------|----------------------------------------------|
| A     | 10              | $\frac{10}{120} \times 360^\circ = 30^\circ$ |
| B     | 20              | $\frac{20}{120} \times 360^\circ = 60^\circ$ |
| C     | 32              | $\frac{32}{120} \times 360^\circ = 96^\circ$ |
| D     | 25              | $\frac{25}{120} \times 360^\circ = 75^\circ$ |
| E     | 15              | $\frac{15}{120} \times 360^\circ = 45^\circ$ |
| F     | 18              | $\frac{18}{120} \times 360^\circ = 54^\circ$ |



Example 2: The Pie-chart drawn below shows how Miss Williams carry out her work.



- Find the value of  $y$
- If the allowance is ₦10,000, how much is spent on food?
- Give the ratio of the miscellaneous to laundry
- What percentage of money is spent on food
- How much is spent for gardening, if allowance is still ₦10,000?

### Solution

i.  $Y + 125^\circ + 65^\circ + 78^\circ = 360^\circ$

$$Y + 268^\circ = 360^\circ$$

$$Y = 360 - 268 = 92^\circ$$

ii. The angle of food =  $125^\circ$

$$\text{Money spent on food} = \frac{125}{360} \times 10000 = \text{N}3,472.22\text{k}$$

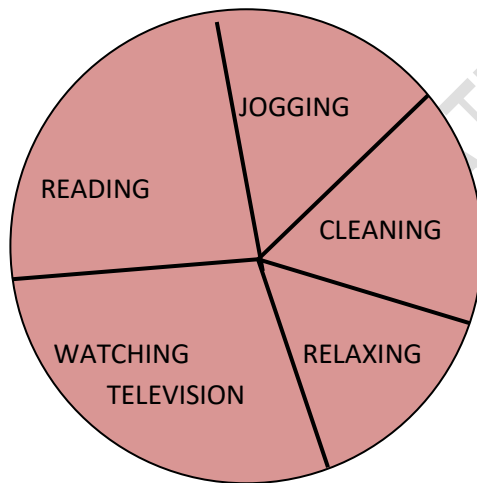
iii. The ratio of miscellaneous to laundry  $92 : 78 = 46 : 39$

iv. The percentage of money spent on food:  $\frac{125}{360} \times 100 = 34.7\%$

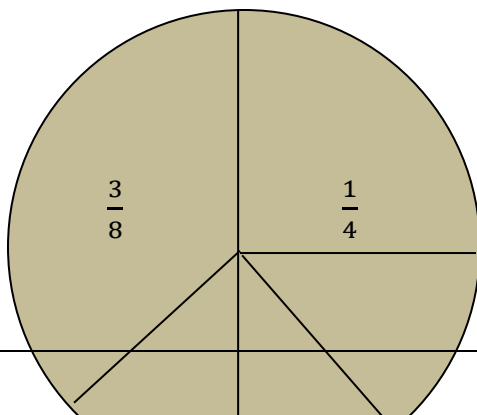
v. How much is spent on Gardening :  $\frac{65}{360} \times 10000 = \text{N}1,805.56\text{k}$

### Evaluation:

1. The pie chart drawn below shows how Hakeem spends his Saturdays.



- Using your protractor, carefully measure each sector
  - How many hours does Hakeem spend watching television?
  - What fraction of time is spent on jogging?
  - Calculate the percentage of time spent on reading
2. Use the pie chart below to answer the following questions



$$\frac{1}{6}$$

x x

- Calculate the angle of each sector
- Find the value of x
- What percentage represents one quarter of the pie?
- What is the ratio in terms of angle  $\frac{1}{6}$  to  $\frac{1}{4}$ ?

**Assignment:** page 190, Ex12, no. 5 -8

### More work on Pie Chart

Example 4: A class of 30 students in grade 11 of a college shows the following:

Yoruba - 28,

Igbo - 32

Hausa - 26

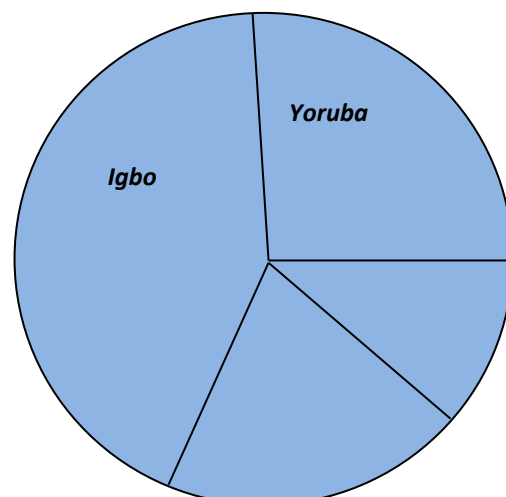
Minority - 24

Show this information on a pie chart

Solution

For Yoruba:  $\frac{8}{30} \times 360^\circ = 96^\circ$

For Igbo:  $\frac{12}{30} \times 360^\circ = 144^\circ$



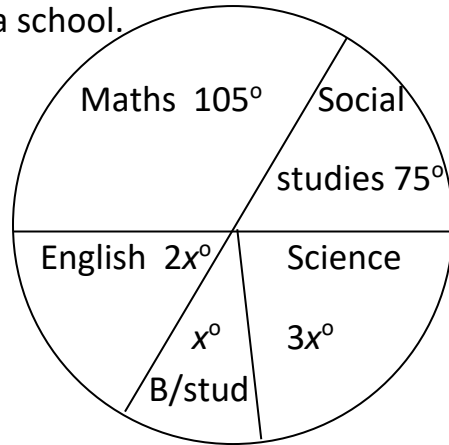
For Hausa :  $\frac{6}{30} \times 360^\circ = 72^\circ$

**Hausa**

For Minority :  $\frac{4}{30} \times 360^\circ = 48^\circ$

**Minority**

Example 5: The pie chart below illustrates the amount of periods allotted the subjects in a school.



- Find the value of  $x$  in degree
- The total number of periods is 24, how many periods is allotted to mathematics?
- What fraction of period allotted to business studies?
- Calculate the percentage period of English.

### Solution

a. The degrees of the subjects :  $105^\circ + 75^\circ + 2x^\circ + x^\circ + 3x^\circ = 360^\circ$

$$180^\circ + 6x^\circ = 360^\circ$$

$$6x^\circ = 360^\circ - 180^\circ$$

$$6x^\circ = 180^\circ$$

Divide both sides by 6

$$x^\circ = 30^\circ$$

b. Total periods = 24; For Mathematics :

$$\frac{105}{360} \times 24 = 7 \text{ periods}$$

- c. The fraction of periods allotted to business studies =  $\frac{30}{360} = \frac{1}{12}$
- d. The percentage periods of English :  $\frac{60}{360} \times 100\% = 16.7\%$

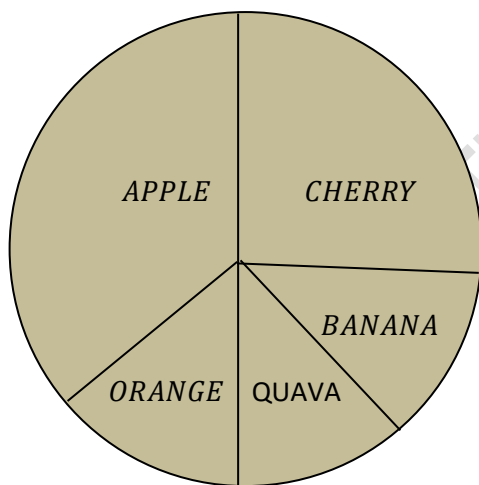
### Evaluation:

1. The table below shows the time spent on private studies by a student in a week on his various subjects.

| Maths | Social studies | English | B/science | Others |
|-------|----------------|---------|-----------|--------|
| 7 hrs | 5 hrs          | 6 hrs   | 2 hrs     | 4 hrs  |

Represent this information on a pie chart.

2. Given that the total fruit is 200. Use protractor to measure the angles and calculate the number of each fruit in a basket represented by the pie chart below.



**Assignment:** page 190, Ex12, no.4,9 10

### More Work :

Example : The table below shows the population of pupils who registered for GCE examination from 2000 to 2005.

| Year           | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 |
|----------------|------|------|------|------|------|------|
| No of students | 180  | 190  | 210  | 220  | 185  | 215  |

Show the information on a pie chart.

**Solution**

Total number of students =  $180+190+210+220+185+215 = 1200$

For Year 2000:  $\frac{180}{1200} \times 360^\circ = 54^\circ$

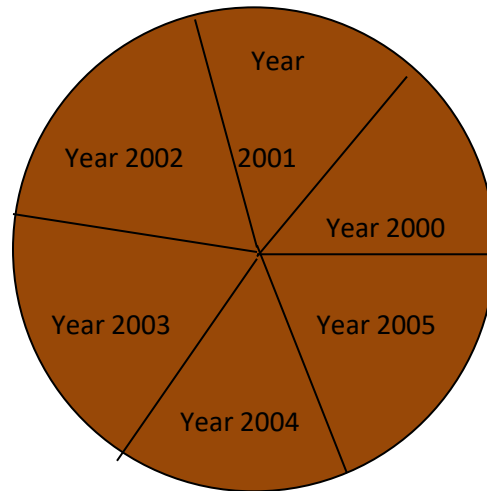
For Year 2001:  $\frac{190}{1200} \times 360^\circ = 57^\circ$

For Year 2002:  $\frac{210}{1200} \times 360^\circ = 63^\circ$

For Year 2003:  $\frac{220}{1200} \times 360^\circ = 66^\circ$

For Year 2004:  $\frac{185}{1200} \times 360^\circ = 55.5^\circ$

For Year 2005:  $\frac{215}{1200} \times 360^\circ = 64.5^\circ$



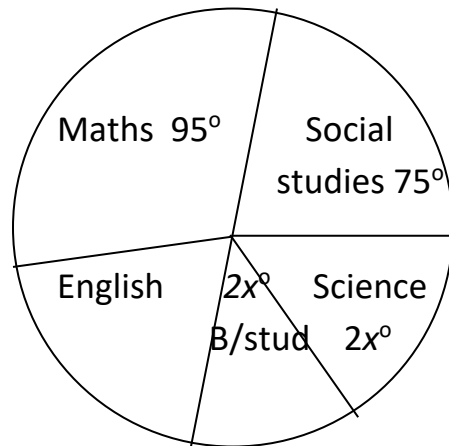
### Usefulness of data for statistical purpose

The usefulness of data for statistical purposes is numerous. Some of them are listed below

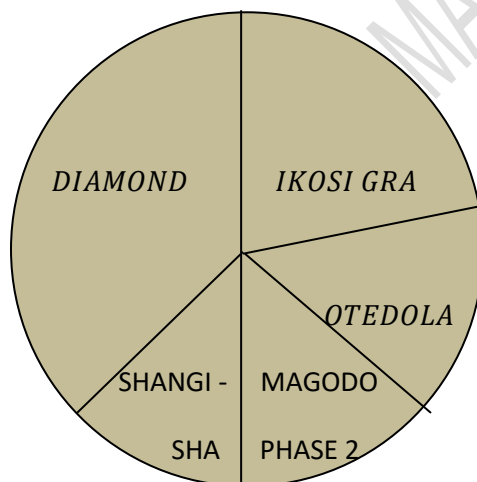
- ✓ to find average of distribution of data
- ✓ to calculate the mid-value of distribution
- ✓ to calculate the range of distribution of data
- ✓ to calculate the number that exist most in a distribution
- ✓ to find percentage and percentile in a distribution
- ✓ can be use to estimate for cost, time or distance in a event
- ✓ to predict outcome of something or possibility of something happening
- ✓ to study the distribution of things in a particular environment
- ✓ to study the relationship between two matters
- ✓ it is used in order to describe something

**Evaluation:**

- a. The pie chart below illustrates the scores of a student in an examination.



- Find the value of  $x$  in degree
  - Calculate the percentage of the student's score in business studies
  - What is the ratio of the score in English to Science
- b. Given that the total number of Toyota Camry 2014 model in the estates shown by the chart is 160. Use protractor to measure the angles and calculate the number of each fruit in a basket represented by the pie chart below.



**Assignment:** page 190, Ex12, no.11, 12 and 13

## WK 9 Probability

**Objectives:** At the end of this lesson students will be able to ;

- Define probability, and all terms associated with probability.

ii. Solve problems on probability

- a. red nor white
- b. green

**Solution:**

a. Total number of balls = 20

Number of white balls = 8

Probability of selecting a **Content**

In everyday activities, we encounter events or experiments whose results we cannot predict in advance. Such experiments are called random experiments.

- Probability is a measure of the likelihood of a required outcome happening.
- An outcome is the result of an experiment in probability.

An event, such as throwing a die, can have different outcomes. For instance, since it has six faces, the probability of throwing a 3 must be one out of six ( $\frac{1}{6}$ ), since there are six different numbers (six possible outcomes).

Therefore, probability is defined in form of a ratio as:

$$\frac{\text{Number of successful/required outcomes}}{\text{Total number of possible outcomes}}$$

Probability can be written as a fraction, decimal or percentage.

**Note:**

1. Probability is always measured as a value between 0 and 1.
2. Probability of an impossible event = 0.
3. *Probability an event occurring + probability of the event not occurring = 1.*

Examples:

1. Out of 100 mangoes, five are bad. What is the probability that a mango chosen at random is good?

Solution:

Total number of mangoes = 100

Number of good mangoes =  $100 - 5 = 95$

Probability of choosing a good mango at random =  $\frac{95}{100} = \frac{19}{20}$

OR

Probability of choosing a good mango at random =  $\frac{5}{100} = \frac{1}{20}$

Probability of choosing a bad mango =  $1 - \frac{1}{20} = \frac{19}{20}$

2. There are 7 red balls, 8 white balls and 5 blue balls in a box. A ball is selected at random from the box. Find the probability that the ball is:

c. white

d. red

e. blue or red

neither white ball =  $\frac{8}{20} = \frac{2}{5}$

b. Probability of selecting a red ball =  $\frac{7}{20}$

c. Probability of selecting a blue ball =  $\frac{5}{20}$

Probability of selecting a red ball =  $\frac{7}{20}$

Probability of selecting a blue or red ball =  $\frac{5}{20} + \frac{7}{20} = \frac{12}{20} = \frac{3}{5}$

d. If the ball is neither red nor white, then it must be blue.

Probability of selecting neither a red nor white ball = Probability of selecting a blue ball

$$= \frac{5}{20}$$

e. There are no green balls.

Probability of selecting a green ball = 0.

Exercises:

New General Mathematics 2, page 206.

Exercise 21c: 5 – 10.

STUDY QUESTIONS:

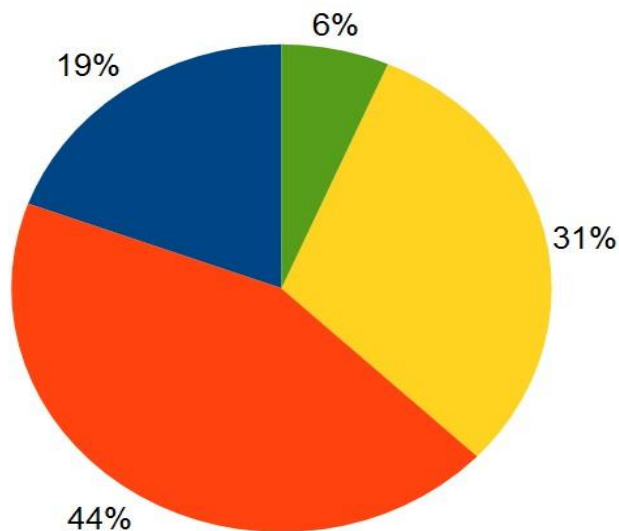
New General Mathematics 2, pages 206 – 208.

Exercise 21c: 11 – 20.

*(Also refer to MAN Mathematics 2, Chapter 18, pages 200 – 208.)*

Class Exercise

1. A bag contains 6 red balls, 5 blue balls and 4 green balls. Find the probability of selecting randomly;
  - I. A green ball
  - II. A red ball
  - III. A blue ball
  - IV. A green or red ball
  - V. A white ball.
  
2. Calculate the respective angles for the percentages shown in the pie chart



SOLUTION.

Total number of balls = 15.  $\frac{31}{100} \times 360 = 111.6$

Number of red balls = 6

Number of green balls = 4

Number of blue balls = 5

$$\text{Pr (green)} = 4/15$$

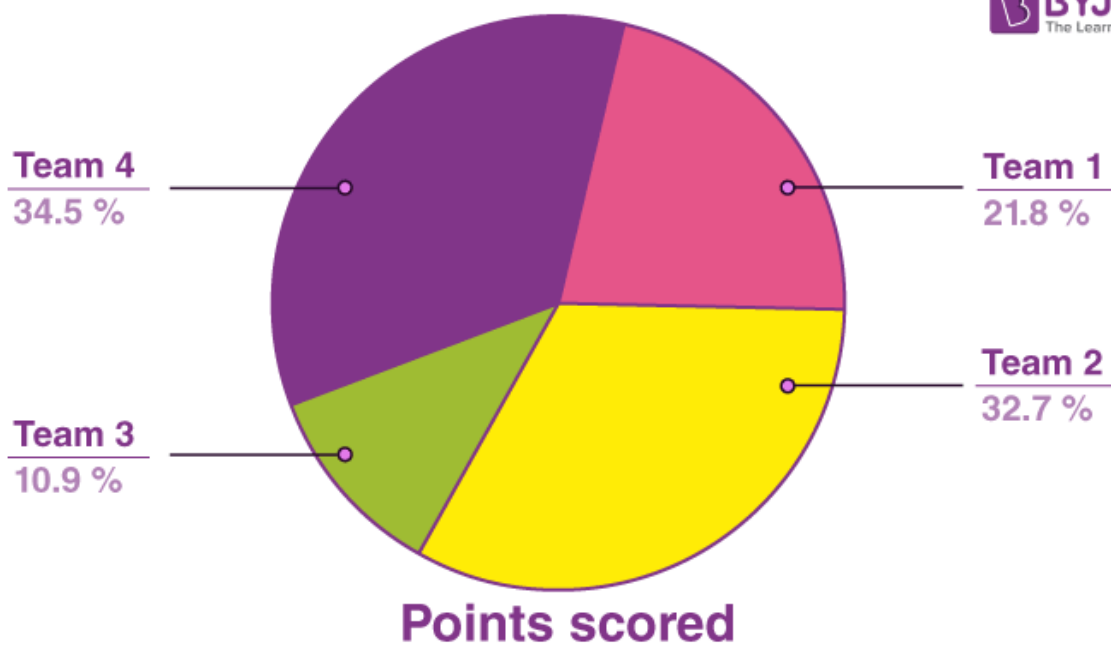
$$\text{Pr (red)} = 2/5$$

$$\begin{aligned}\text{Pr (green or red)} &= P\left(\frac{4}{15} + \frac{2}{5}\right) \\ &= \frac{4+6}{15} = \frac{10}{15} = \frac{2}{3}\end{aligned}$$

Additional questions.....

The following are the ages of 16 college students: 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30.

- 1) What is the range of ages at the college?
  - A. 16
  - B. 18
  - C. 28
  - D. 22
  - E. None of the above
- 2) What is the median age of the students at this college?
  - A. 16
  - B. 18
  - C. 22
  - D. 26
  - E. None of the above
- 3) The mean of the ages is ?
  - A. 360
  - B. 350
  - C. 450
  - D. 230
- 4) There are six sides to a standard die, numbered 1-6. What is the probability of NOT getting a factor of 6?
- 5) Find the mode of the call received on 7 consecutive day 11,13,13,17,19,23,25
- 6) Find the median of the call received on 7 consecutive days 11,13, 17, 13, 23,25,19
- 7) A \_\_\_\_\_ is a bar graph that shows data in intervals.
- 8) The point (4,0) lies on which of the following?



9) Which team has the highest score?

10) What is the average score of all the teams?